

Section 78 — Master Suffix Structures Mixed Contest

Contest contract

Six new problems mix suffix ordering, LCP connectivity, suffix-automaton path weights, cross-string propagation, and palindromic occurrence aggregation. They are not restatements of Sections 76–77.

Suggested order: A, D, E, B, C, F. Budget about five hours, then write a proof before reading the editorial.

Coverage map

Problem	Main pressure point
A	eliminate cyclic-rotation candidates in linear time
B	process every occurrence threshold through LCP edges and DSU
C	distinguish a substring from its many occurrences in SAM path DP
D	propagate the best match through suffix links
E	propagate palindrome occurrence counts
F	combine per-string SAM evidence with an order statistic

Post-contest questions

For each problem, identify the exact object represented by one state, the direction in which information must flow, why every candidate is counted, and why no candidate is counted twice.

Technique recognition

Evidence	Structure or ordering
compare all cyclic rotations	candidate elimination / Booth
answer changes as an LCP threshold decreases	sort LCP edges and use DSU
lexicographic substring weighted by occurrences	weighted SAM path blocks
evidence gathered while scanning another string	suffix-link propagation
aggregate palindrome frequencies	longer-to-shorter ctree links
a condition must hold in many strings	per-string evidence plus an order statistic

Transitions flow from shorter strings to extensions. Occurrence and matching evidence usually flows in the opposite direction through suffix links. Decide which quantity you have before choosing the processing order.

Discovery walkthrough: Repetition Spectrum

Treat suffix-array positions as vertices. Adjacent suffixes connected by an LCP value w share a prefix of length w . Process these adjacency edges in decreasing w . After all edges with weight at least w are added, a DSU component is exactly a maximal suffix-array interval whose suffixes share a prefix of length at least w .

If its size is s , that prefix occurs at least s times. Record the best length when components merge, then take suffix maxima over occurrence counts. This transforms interval minima into threshold connectivity—a pattern also useful outside strings.

Discovery walkthrough: Multi-Archive Commonality

Build one SAM for the base archive. While scanning another archive, record the best match length reaching each state. A state also represents shorter strings, so propagate evidence through suffix links and clamp it by the suffix state's `len`. Only after one archive has correct evidence should it contribute to the cross-archive order statistic.

The crucial separation is:

```
scan characters -> repair evidence along suffix links -> combine archives
```

Combining before propagation undercounts substrings represented by suffix states that were never the final streaming state.

Proof and implementation checklist

- Does one state count distinct strings, occurrences, or end positions?
- Is DP ordered along transitions or against suffix links?
- Are multiple occurrences deliberately weighted or deliberately collapsed?
- Are per-string arrays cleared only on touched states?
- Are path counts capped before addition can overflow?
- Does cyclic comparison avoid constructing every complete rotation?

Independent oracle map

For short strings, explicitly list rotations, suffixes, every distinct substring, every occurrence, and every palindromic interval. Sort these literal lists to answer A–F. Such oracles share no rank, automaton, suffix-link, or eertree logic with the optimized solutions and therefore expose mistaken state semantics rather than reproducing them.

Exercises

Problems A–F in this section form the complete mixed contest.